Grade 10 – Term 1
Decision Trees with Staged Filtering: C9 Practice

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Evaluated criteria

C9 Constructs and interprets decision trees for staged filtering rules, applying decision criteria in the specified order to identify the final alternative.

1 Introduction

The summary tables provide four statistics for each alternative, all in thousand USD and rounded to two decimal places. Use these supplied statistics directly; you are not required to recalculate them from the payoff matrices.

- **Maximax:** select the highest Max Payoff (the alternative's maximum payoff).
- **Maximin:** select the highest Min Payoff (the alternative's minimum payoff).
- **Minimax Regret:** select the lowest Max Regret. Regret in each state is the best payoff among all six original alternatives in that state minus the alternative's payoff; Max Regret is the largest of these regrets.
- **EMV (Expected Monetary Value):** select the highest EMV, calculated using the stated probabilities as payoff weights.

Apply each problem's filtering rules sequentially. Compare only alternatives retained from the preceding stage; eliminated alternatives cannot reenter. All original summary statistics remain fixed: do not recalculate any statistic after eliminating alternatives. When asked to keep the best k alternatives, retain every alternative tied at the cutoff, even if more than k remain. At the final stage, report a tie explicitly if multiple alternatives share the best value.

Represent each process in a four-column filtering table: # Step, Description, Left Alternatives, and Eliminated Alternatives. Step 0 lists all six alternatives. For each subsequent stage, record the criterion, filtering rule, relevant numerical values, and alternatives retained or eliminated at that stage. Identify the final selected alternative (or tied alternatives) in the last row. The problems use two to four filtering stages and vary the criteria applied.



2 Decision Trees with Staged Filtering

Problems:

2.1. City logistics policy

2.2. Regional distributor expansion

2.3. Trade finance guarantees

2.4. Insurance portfolio mixes

2.1 City Logistics Policy

Problem Description

A city logistics office must choose one policy among $B_1, B_2, B_3, B_4, B_5, B_6$. Payoffs (thousand USD) and probabilities are:

State	Probability	B_1	B_2	B_3	B_4	B_5	B_6
T_1	0.50	95	90	110	85	100	92
T_2	0.30	70	75	60	80	65	78
T_3	0.20	30	40	10	55	35	20

The summary statistics below are already calculated. Use them directly; do not recompute them from the payoffs. All summary values are in thousand USD, to two decimal places.

Alternative	EMV	Max Payoff	Min Payoff	Max Regret
B_1	74.50	95.00	30.00	25.00
B_2	75.50	90.00	40.00	20.00
B_3	75.00	110.00	10.00	45.00
B_4	77.50	85.00	55.00	25.00
B_5	76.50	100.00	35.00	20.00
B_6	73.40	92.00	20.00	35.00

Filtering rules (all values in thousand USD; retain ties at every cutoff): (1) EMV: keep the 4 highest EMV values. (2) from those, Maximin: keep the 2 highest Min Payoff values. (3) from those, Minimax Regret: select the lowest Max Regret value.

Questions/Tasks



C9 Apply the filtering rules using the supplied results. Construct a step-by-step filtering table showing the rule, relevant numerical values, and alternatives retained and eliminated at each stage, including the final decision.

Solution

# Step	Description	Left Alternatives	Eliminated Alternatives
0	Initial alternatives	$\{B_1, B_2, B_3, B_4, B_5, B_6\}$	—
1	EMV: keep the 4 highest EMV values.	$B_4(77.50)$, $B_5(76.50)$, $B_2(75.50)$, $B_3(75.00)$	$B_1(74.50)$, $B_6(73.40)$
2	Maximin: keep the 2 highest Min Payoff values.	$B_4(55.00)$, $B_2(40.00)$	$B_5(35.00)$, $B_3(10.00)$
3	Minimax Regret: select the lowest Max Regret value.	**Selected:** $B_2(20.00)$	$B_4(25.00)$



2.2 Regional Distributor Expansion

Problem Description

A regional distributor must choose one expansion plan among $X_1, X_2, X_3, X_4, X_5, X_6$. Payoffs (thousand USD) and probabilities are:

State	Probability	X_1	X_2	X_3	X_4	X_5	X_6
U_1	0.30	130	120	140	110	125	118
U_2	0.25	90	95	80	100	88	92
U_3	0.25	60	70	55	85	65	75
U_4	0.20	20	40	10	60	45	35

The summary statistics below are already calculated. Use them directly; do not recompute them from the payoffs. All summary values are in thousand USD, to two decimal places.

Alternative	EMV	Max Payoff	Min Payoff	Max Regret
X_1	80.50	130.00	20.00	40.00
X_2	85.25	120.00	40.00	20.00
X_3	77.75	140.00	10.00	50.00
X_4	91.25	110.00	60.00	30.00
X_5	84.75	125.00	45.00	20.00
X_6	84.15	118.00	35.00	25.00

Filtering rules (all values in thousand USD; retain ties at every cutoff): (1) Maximax: keep the 3 highest Max Payoff values. (2) from those, EMV: select the highest EMV value.

Questions/Tasks



C9 Apply the filtering rules using the supplied results. Construct a step-by-step filtering table showing the rule, relevant numerical values, and alternatives retained and eliminated at each stage, including the final decision.

Solution

# Step	Description	Left Alternatives	Eliminated Alternatives
0	Initial alternatives	$\{X_1, X_2, X_3, X_4, X_5, X_6\}$	—
1	Maximax: keep the 3 highest Max Payoff values.	$X_3(140.00)$, $X_1(130.00)$, $X_5(125.00)$	$X_2(120.00)$, $X_6(118.00)$, $X_4(110.00)$
2	EMV: select the highest EMV value.	**Selected:** $X_5(84.75)$	$X_1(80.50)$, $X_3(77.75)$



2.3 Trade Finance Guarantees

Problem Description

A trade finance agency chooses one guarantee policy among $G_1, G_2, G_3, G_4, G_5, G_6$. States and probabilities are shown below. Payoffs are in thousand USD. Policies G_4 and G_5 balance upside and downside because they diversify by sector, while G_3 is highly cyclical.

State	Probability	G_1	G_2	G_3	G_4	G_5	G_6
M_1	0.45	100	94	114	90	104	96
M_2	0.35	72	78	64	84	70	80
M_3	0.20	32	44	12	56	38	26

The summary statistics below are already calculated. Use them directly; do not recompute them from the payoffs. All summary values are in thousand USD, to two decimal places.

Alternative	EMV	Max Payoff	Min Payoff	Max Regret
G_1	76.60	100.00	32.00	24.00
G_2	78.40	94.00	44.00	20.00
G_3	76.10	114.00	12.00	44.00
G_4	81.10	90.00	56.00	24.00
G_5	78.90	104.00	38.00	18.00
G_6	76.40	96.00	26.00	30.00

Filtering rules (all values in thousand USD; retain ties at every cutoff): (1) EMV: keep the 5 highest EMV values. (2) from those, Maximax: keep the 4 highest Max Payoff values. (3) from those, Maximin: keep the 2 highest Min Payoff values. (4) from those, Minimax Regret: select the lowest Max Regret value.

Questions/Tasks



C9 Apply the filtering rules using the supplied results. Construct a step-by-step filtering table showing the rule, relevant numerical values, and alternatives retained and eliminated at each stage, including the final decision.

Solution

# Step	Description	Left Alternatives	Eliminated Alternatives
0	Initial alternatives	$\{G_1, G_2, G_3, G_4, G_5, G_6\}$	—
1	EMV: keep the 5 highest EMV values.	$G_4(81.10)$, $G_5(78.90)$, $G_2(78.40)$, $G_1(76.60)$, $G_6(76.40)$	$G_3(76.10)$
2	Maximax: keep the 4 highest Max Payoff values.	$G_5(104.00)$, $G_1(100.00)$, $G_6(96.00)$, $G_2(94.00)$	$G_4(90.00)$
3	Maximin: keep the 2 highest Min Payoff values.	$G_2(44.00)$, $G_5(38.00)$	$G_1(32.00)$, $G_6(26.00)$
4	Minimax Regret: select the lowest Max Regret value.	**Selected:** $G_5(18.00)$	$G_2(20.00)$



2.4 Insurance Portfolio Mixes

Problem Description

An insurance broker considers six portfolio mixes Y_1 to Y_6 under four climate-risk states with probabilities W_1:0.30, W_2:0.25, W_3:0.25, W_4:0.20. Payoffs are in thousand USD. Y_4 is conservative and protects downside in severe-weather years, while Y_3 has the highest upside but a weak stress outcome.

State	Probability	Y_1	Y_2	Y_3	Y_4	Y_5	Y_6
W_1	0.30	136	126	146	116	130	121
W_2	0.25	94	98	84	103	90	95
W_3	0.25	66	74	58	88	69	77
W_4	0.20	28	46	14	64	48	38

The summary statistics below are already calculated. Use them directly; do not recompute them from the payoffs. All summary values are in thousand USD, to two decimal places.

Alternative	EMV	Max Payoff	Min Payoff	Max Regret
Y_1	86.40	136.00	28.00	36.00
Y_2	90.00	126.00	46.00	20.00
Y_3	82.10	146.00	14.00	50.00
Y_4	95.35	116.00	64.00	30.00
Y_5	88.35	130.00	48.00	19.00
Y_6	86.90	121.00	38.00	26.00

Filtering rules (all values in thousand USD; retain ties at every cutoff): (1) EMV: keep the 3 highest EMV values. (2) from those, Minimax Regret: keep the 2 lowest Max Regret values. (3) from those, Maximin: select the highest Min Payoff value.

Questions/Tasks

C9 Apply the filtering rules using the supplied results. Construct a step-by-step filtering table showing the rule, relevant numerical values, and alternatives retained and eliminated at each stage, including the final decision.

Solution

# Step	Description	Left Alternatives	Eliminated Alternatives
0	Initial alternatives	$\{Y_1, Y_2, Y_3, Y_4, Y_5, Y_6\}$	—
1	EMV: keep the 3 highest EMV values.	$Y_4(95.35)$, $Y_2(90.00)$, $Y_5(88.35)$	$Y_6(86.90)$, $Y_1(86.40)$, $Y_3(82.10)$
2	Minimax Regret: keep the 2 lowest Max Regret values.	$Y_5(19.00)$, $Y_2(20.00)$	$Y_4(30.00)$
3	Maximin: select the highest Min Payoff value.	**Selected:** $Y_5(48.00)$	$Y_2(46.00)$